

Design and Development of Real Time Indoor and Outdoor Air Quality Monitoring System Based on IoT Technology

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Abstract— While the countries grow to be developed and industrial, the pollution level is increased significantly, and this becomes a critical downfall for the people's health. This paper presents a system of smart air quality monitoring for urban city area by using IoT Technology. The main purpose of this research is to minimize, improvise, and use low-cost configuration, yet with equal ability to perform as industrial sensor capability, as required by the IEEE 1451 Sensor Standard. This enables the usage of this system in range of 20 meters for the WIFI module. This module consists of two sensors that function to detect concentrate elements in nearby environment. The first sensor is an air quality sensor that functions to detect the value of carbon dioxide in the environment. The second sensor detects and indicates the temperature and humidity. Both sensors will show the values detected through wireless monitor, either by smartphone or any device that can access web-based or application platforms. This is to ensure that the value is detected, hence ability to observe. Any malfunction can be reset remotely, and the user can determine the next decisions. The proposed system, by combination of esp32, IoT Platform and the Wi-Fi modules, is also able to give immediate data reading, while being remotely observed by notice from the system output.

Keywords—Smart air quality, IoT Technology, Real Time Monitoring, Temperature, Humidity

I. INTRODUCTION

This Pollution has long been a worldwide topic among specialists. Air pollution is on the affects both indoor and outdoor environment. Excessive emission of pollutant into the air will affect human health. The conventional method to detect air pollution detection is simply by using photos, and then manually determining the concentration of haze in atmosphere. In order to ensure the best accuracy, reliability

and minimalistic system for detecting the concentration level of air pollution, research and product testing have to made to support the requirement of collecting the value of air pollution, besides achieving the full efficiency to remotely access data measurement in the system[1].

There have been several studies related to Air pollution as the pollution has long been a worldwide topic among specialists. In [1], the author developed a smart air pollution monitoring system that has sensors to detect carbon monoxide, smoke and gas, and has an LCD. The LCD displays the current reading for users in real time. However, the data collected by the system can be printed for can only be viewed every 8 hours. This project adapts this setup configuration for alert system that will give feedback to the users about health level of the sample location; measured by the sensor. However, this system does not support internet of things and data retention. This will void the air pollution data, due to limited time for data retrieval, thus preventing the user to revise the value that had been collected.

The feature in [2] adapted the configuration from [1]; which abides the regulation determined by the IEEE standard regulation for specification grading, by monitoring air pollution developed in the sample environment. The system detects gases such as CO, NO₂, SO₂ and O₃ using semiconductor sensors. However, some of the gases need to be recalibrated before starting to take measurement for each type of gases to ensure having accurate value for each element data. The proposed configuration utilizes low-cost sensor version

Author in [3] reviewed a model which is a suitable and potential option to implement in real-time for air pollution detection and monitoring, which uses microcontroller and WIFI module. In pollution monitoring, there are various

advantages and disadvantages of existing methods such as electrochemical gas sensing method, tapered element oscillating micro balance (TEOM), BETA – Attenuation method (BAM), black smoke method, light scattering method, and WSN-based air pollution monitoring systems. By comparing these methods, the most effective method can be achieved by combination to detect air pollution accurately. The best selection should be the one having the most advantages. User should configure the best options to effectively detect air pollutants.

The system in [4] has the option for upgrading the system to support internet of things. The system senses the air quality, gas, temperature, and humidity from given sample. The system can detect carbon dioxide, nitrogen oxide, temperature, and humidity concentration in the industrial area. Visual output is shown on LCD and web-based system. However, this system is rather bulky, and requires high cost for the design, since it also takes the measurement of other elements in the environment.

Author in [5] designed a system to measure air quality and volume concentration based on parts per million (PPM) metrics. The data obtained by the sensors are analyzed using is transmitted to mobile device to show the user the output value from the sensor. This system is made to connect to mobile device to view the value, but its configuration is inefficient for the user to view the value taken.

The author in [6] proposed a cost-efficient system for air monitoring detection. The detected value will be evaluated manually by the user, following the standard regulation of air quality. This system also lacks the integration of internet of things feature to remotely present visual output to customers.

The aim of this research is to design and development of real time indoor and outdoor air quality monitoring system based on IoT technology. Moreover, this paper proposed a method for measuring the value of carbon dioxide element, ambient temperature, and humidity in environment samples by using MQ135 sensor, DHT22 sensor, and ESP32 Microcontroller unit as interface. An additional system has been implemented which a remote reset for the sensor. This feature is an optional configuration for the users if they want to maintain the sensor if some sensors malfunction or failed during the operation of taking samples.

II. DESIGN OF SMART INDOOR AND OUTDOOR AIR QUALITY MONITORING SYSTEM USING MQ135 SENSOR

The block diagram for the proposed design is shown in Figure 1. This system has one microcontroller which is the ESP32 MCU, and sensors to sense either indoor or outdoor air quality, with temperature and humidity. Upon detection, the system is configured to present the data through an Internet of Things application, which is BLYNK, and the final result is sent to the user for observation.

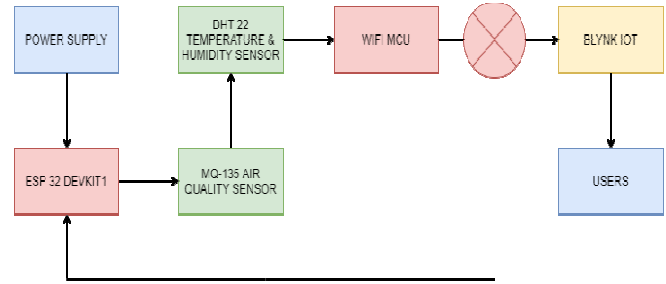


Fig. 1. Block diagram of the proposed system

As shown in the circuit diagram (Figure 2), the input power supplies to operate the entire system are 3.3 Volts and 5 Volts. The process of the proposed system starts from the environment particle sample, where there is presence of CO₂ with temperature and humidity. The microcontroller, ESP32 DEVKIT1 configures the entire system that has been integrated with the microcontroller. After the configuration setup, the ESP32 DEVKIT1 configures the sensor to process the value that it collects from the sensor to calculate the concentration of CO₂ with temperature and humidity, in percentage, present in the environment. After data sensing has been configured, the data will be autonomously transferred to Wi-Fi MCU to process the data from both sensors to transfer value that it obtains to the BLYNK application. The output of all these numbers will be given to the users, which can be seen from BLYNK application[7-8].

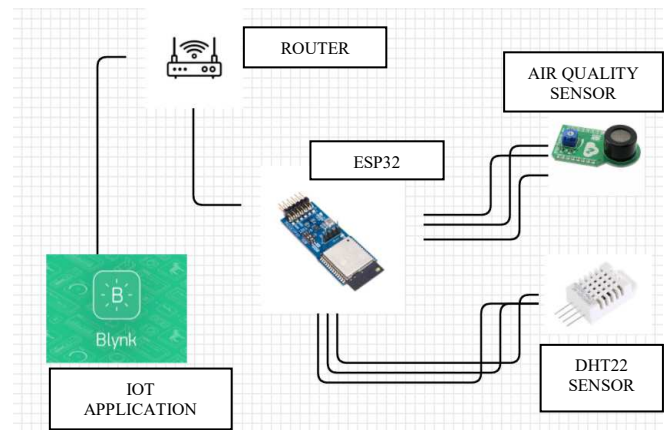


Fig. 2. Circuit diagram of Smart Indoor and Outdoor Air Quality Monitoring System using MQ135 Sensor

III. RESULT AND DISCUSSION

The Real Time Indoor and Outdoor Air Quality Monitoring System had been tested to validate its efficiency. This part of study focused on gathering the final output from the ESP32 to be shown in BLYNK app. The BLYNK app was used to observe or monitor the values from the sensors, obtained from indoor and outdoor ambient samples. The indoor sample was then compared to the standard level of air quality, which the air quality index (AQI) level, as shown in Table 1.

TABLE I. INDOOR CARBON DIOXIDE AIR QUALITY INDEX [9] AND [10]

Carbon Dioxide (CO ₂) PPM	Air Quality Level Index (AQI)
2500 or more	Bad for health

1500 below	Mediocre and air is filthy
1000 below	Fair & good
600 below	Excellent

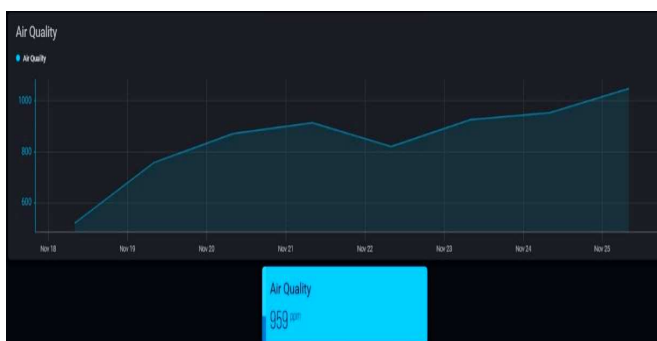
The part of data gathering was by collecting sample of indoor ambient for one week. As shown in Figures 3 (a) – (c), the calculated sample for the one-week data included the parameters of humidity, temperature, and air quality observed. The figures also show the value range and the average of temperature, humidity, and air quality. From the figures, it can be seen that the temperature ranged from 20° Celsius to 35° Celsius. The ambient humidity ranged from 40% - 60%, and had average of 50% throughout the whole one week. The Air quality value for a week in this project is at average value of 959 Parts per Million.



(a)



(b)



(c)

Fig. 3. One Week Indoor Results: (a)Temperature, (b)Humidity, and (c)Air Quality

The change of temperature and humidity values was influenced by air quality. This was also due to influence of odor intensity of air [11] present in the room environment. According to [9] and [10], good air quality index for indoor ranges from 350-1000 PPM, which is the safe operating level

of CO₂. This can be seen from the different values of temperature and humidity as shown in Figures 3 and 4, from different days in the whole one-week samples. This finding agrees with the fact from [12-15], that the value of other particles present in the environment can contribute to different value of air quality in indoor environment, which also affects the wellness of a person by certain level of CO₂ concentration in indoor ambient.

The second part of data gathering focused on collecting outdoor ambient sample for one week, as shown in Figures 4 (a) – (c). The monitoring application has both parameters and values from both sensors. Humidity during the one week ranged from 55% to 75%, with average value of 68%, while the temperature ranged from 28° Celsius to 31° Celsius, with average value of 30° Celcius. The one-week air quality result ranged from 700PPM to 1100PPM, with average of 823PPM.



(a)



(b)



(c)

Fig. 4. One Week Outdoor Results: (a)Humidity, (b)Temperature, and (c)Air Quality

There were several factors that led to the different values of temperature and humidity, with relation to air quality collected from the outdoor environment sample such as wind speed, wind direction, temperature, humidity, solar radiation,

and rainfall [11]. As mentioned in rainfall [11], these factors will influence the concentration of pollutants in the study area. This can be seen in Figure 4, which shows different value of air quality with temperature and humidity in outdoor environment samples.

Figure 5 (a) and (b) show the final prototype of proposed design. The real time indoor and outdoor air quality monitoring system is placed in a box that is designed to protect and avoid any natural element that can damage the components inside, such as water and dust.

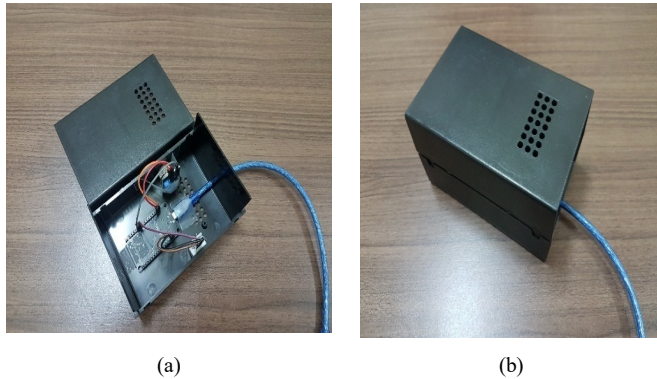


Fig. 5. (a) Overall Circuit Setup and (b) Final Design for smart indoor and outdoor air quality monitoring system using MQ135 sensor

IV. CONCLUSION

Nowadays, the safety is playing a critical part that make implementing of reliable safety system necessary matter in different places. While the countries grow to be developed and industrial, Air pollution is receiving extensive attention as it can jeopardize people's health. In this study, real time indoor and outdoor air quality monitoring system based on IoT technology was successfully designed, built and tested. In this system, MQ135 sensor, managed by ESP32 microcontroller was used. The microcontroller, ESP32, and both MQ135 and DHT22 sensors have proven their reliability for use in the system. The implementation of Internet of Things (IoT) enables remote monitoring and viewing of the average value by the user for observation making as the warning and notification is delivered through a smart phone application.

ACKNOWLEDGMENTS

The authors would like to thank Management and Science University and Ministry of Higher Education Malaysia for providing the financial support for this research.

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